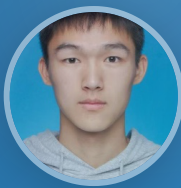


Wang Shaohui

WANGSHAOHUI

Human Factors Experiment & Data Analysis Engineer



A data-driven testing engineer with a **Master's in Sports Science** and hands-on experience in **wearable sensor data collection, algorithm validation, and experimental data analysis**. Seeking to bring my cross-disciplinary expertise in **human physiology, data science, and standardized testing** to TUV Rheinland's Human Factors team.

Email: 2194712714@qq.com

Phone: +86 176-3964-7964 (WeChat)

Location: Shenzhen, China | Open to relocation

English: CET-6 | Professional English: 94/100

Human Factors Testing

Sensor Data Collection

Experimental Data Analysis

Algorithm Evaluation

Python & R

Statistical Modeling



Education

EDUCATION

Zhengzhou University — Master of Sports Science (Sports Training)

GPA: **3.89 / 4.3** | Research Focus: Physical Health Promotion, Sports Performance, Sports Data Analysis

Sep 2023 – Jun 2026

- **Core Coursework:** Sports Training Theory & Methods (96), Sports Pedagogy (93), Professional English (94)
- Systematically trained in **experimental design, data collection, and statistical analysis** for human performance research
- Conducted multiple experiments involving **human subjects** — designing test protocols, collecting physiological & behavioral data, and analyzing results

Zhengzhou University — Bachelor of Physical Education

Key Courses: Sports Training (90), Motor Learning & Control (90), Basketball (93)

Sep 2019 – Jun 2023



Work Experience

EXPERIENCE

Suunto — Algorithm Test Engineer

Jul 2026 – Present

- **Human-Centered Data Collection:** Designed and executed test protocols for heart rate, SpO2, sleep staging, motion pattern, and barometric sensor algorithms — collecting physiological and behavioral data from human participants under controlled conditions
- **Standardized Testing Process:** Built end-to-end standardized test workflows covering test case design, data collection protocols, annotation guidelines, and quality verification for wearable device algorithms



Skills

SKILLS



Human Factors Data Collection

HR, SpO2, sleep, motion & barometric sensor data — protocol design, participant testing, annotation guidelines



Data Analysis & Statistics

Python (scikit-learn), R (tidyverse, meta/metafor), SPSS, Stata — statistical modeling, meta-analysis, effect size computation



Algorithm Evaluation

Accuracy, recall, F1, Bland-Altman agreement, kappa coefficient — quantitative performance assessment framework



AI & Workflow Automation

AI agent integration for test case generation, defect reporting, and process orchestration



Human Performance Domain

Deep understanding of human physiological signals, exercise physiology, and motor behavior — bridging human factors and product testing



Tools & Workflow

Git version control, Office suite, technical report writing in English & Chinese



Additional Experience

PRACTICE



Basketball Referee (Level 1)

- **Experimental Data Analysis:** Performed quantitative analysis on real-world human data to evaluate algorithm performance — identifying and diagnosing defects in HR detection, sleep staging, and motion recognition algorithms
- **Algorithm Evaluation:** Established performance evaluation framework comparing algorithm versions using accuracy, recall, F1-score, Bland-Altman agreement, and kappa coefficient metrics
- **Cross-Functional Collaboration:** Worked with algorithm and product teams to define acceptance criteria and quantify algorithm performance improvements across iterative versions
- **AI-Powered Workflow:** Integrated AI agents into testing workflows — automating test case generation, defect report compilation, and process orchestration

Certified first-level referee — strong observation, decision-making, and communication skills



Sports Science Tutoring

Tutored peers in data analysis and academic writing — cross-disciplinary knowledge integration



Physical Education Teacher

Part-time PE teacher during undergraduate years — communication and organizational skills



Publications

PUBLICATIONS

Effects of three strength training methods on lower extremity strength, jump and sprint performance: A network meta-analysis

Frontiers in Physiology **JCR Q2** — Built a network meta-analysis model comparing three training interventions using standardized effect sizes and heterogeneity testing

Combined training outperforms single-form in improving lower extremity reactive strength index: A network meta-analysis

Journal of Science in Sport and Exercise **JCR Q3** — First systematic comparison of combined vs. single plyometric training on RSI

▸ Conference Presentations: National Sports Science Congress, National Competitive Sports Science Congress, Asia-Pacific Sports & Exercise Science Congress



Research Projects

PROJECTS

❖ MOE Humanities & Social Sciences Project

Research on physical health crisis and mitigation strategies for college students under the Healthy China initiative

❖ Henan Provincial Philosophy & Social Sciences Planning Project

Network meta-analysis of physical health interventions and RE-AIM evaluation of health behaviors

❖ MOE Industry-Education Cooperation Project

Strategies for improving student physical health under sports-education integration

❖ University-Level Projects

Seven first-class courses and educational reform projects at Zhengzhou University